

Te-Hsun Tsai

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EDUCATION

University of Washington

Master of Science in Statistics (Advanced Methods & Data Analysis) (GPA: 3.88/4.0)

Ph.D.-level Pass — UW Statistics Theory Exam

Seattle, WA | Sep. 2025 – Expected Mar. 2027

National Chengchi University

Bachelor of Science in Statistics (GPA: 3.88/4.0)

Taipei, Taiwan | Sep. 2021 – Jun. 2025

RESEARCH INTERESTS

Reliable & Generalizable ML | Machine Learning for Sequential & Multimodal Data | Robustness, Distribution Shift & Uncertainty

PUBLICATIONS

Tsai, T.-H., Chien, F. P., Li, V. C.-S., et al. “Decoding Prior-Induced Speech States from fMRI under Identical Acoustics.” *Submitted to IEEE ICASSP 2027*, 2026.

RESEARCH EXPERIENCE

Research Intern — Machine Learning for Speech & Neural Data

Academia Sinica | Advisor: Prof. Yu Tsao

Taipei, Taiwan

Jun. 2026 – Sep. 2026

Prior-Induced Speech-State Decoding — ICASSP 2027

- Developed Wave-4 S5, combining local position-preserving voxel representations with bidirectional state-space modeling for high-dimensional, small-sample fMRI decoding under acoustically identical speech stimuli.
- Built reproducible evaluation and model-comparison pipelines spanning SVM, residual MLP, Transformer, and S5 models with fold-specific preprocessing and multiple-comparison-corrected statistical testing.
- Achieved 64.29% held-out Pre/Post balanced accuracy and up to 68.45% Clean-to-Post projection, demonstrating cross-condition generalization to unseen Clean responses.
- Linked Clean-to-Post projection to sentence-level comprehension gains (Spearman ρ up to 0.632, BH-FDR $q = 0.0014$), providing behavioral validation of the learned neural-state boundary.

Speech-Brain Representation Alignment

- Extracted hidden-state representations from pretrained speech encoders (WavLM, Wav2Vec2, Whisper) and language models (GPT-2, Phi-4), and quantified their alignment with EEG and fMRI responses across clean and noisy speech conditions.
- Evaluated layer-wise model-brain alignment using ridge regression, leave-run-out cross-validation, Pearson correlation, noise-ceiling normalization, and robust subject- and channel-level aggregation.
- Compared raw and noise-ceiling-normalized alignment across model families, Clean/Noisy conditions, and all-channel, alpha-band, visual, and semantic neural subsets.
- Investigated representation robustness across model families, signal conditions, and neural modalities, identifying layers and representations that generalized across heterogeneous settings.

Research Assistant — Cause-of-Death Modeling

University of Washington | Research Supervisor: June Yang, Research Scientist

Seattle, WA

Sep. 2025 – Dec. 2025

- Developed machine-learning benchmarks using SVM and Random Forest for automated cause-of-death classification across heterogeneous verbal-autopsy datasets with highly imbalanced, long-tailed labels.
- Designed hierarchical classification architectures with soft pruning and chapter-to-detail constraints, and performed systematic hyperparameter tuning across age-stratified cohorts.
- Built strict no-leakage evaluation and multimodal feature-fusion pipelines for heterogeneous clinical data, emphasizing robust evaluation under sparse and small-sample settings.

Undergraduate Researcher — Forecasting & Data Assimilation for Chaotic Systems

National Center for Theoretical Sciences (NCTS) | Advisor: Prof. Feng-Nan Hwang

Taiwan

Jul. 2024 – Aug. 2024

- Developed hybrid forecasting methods combining LSTM/ARIMA predictors with Kalman-filter data assimilation for nonlinear Lorenz-63 dynamics.
- Conducted 200+ controlled experiments evaluating forecast robustness across training-data size, forecast horizon, observation noise, and model misspecification.

GRADUATE COURSE PROJECT

Adaptive Conformal Prediction for Chaotic Time Series

- Studied split conformal prediction under closed-loop chaotic forecasting, demonstrating substantial coverage degradation as forecast horizon increased.
- Evaluated rolling-window and EWMA adaptive calibration strategies, recovering near-nominal long-horizon coverage while characterizing the coverage-interval-width trade-off.

PROFESSIONAL EXPERIENCE

Optimization & Scheduling System Engineer

Elementary Schools

Taipei, Taiwan

Jan. 2025 – Jun. 2025

- Developed and deployed an integer linear programming scheduling system optimizing 152 course-teacher assignments under policy and availability constraints, reducing administrative workload by 80%+ through automated validation and FET-compatible workflows.

TECHNICAL SKILLS

Programming & Tools: Python, R, SQL, MATLAB, C++, SAS, Stata, LaTeX, Git

Machine Learning & Reliability: PyTorch, Transformers, S5/Mamba, SVM, Random Forest, XGBoost, Conformal Prediction, LLM Evaluation

Sequential & Neural Data: LSTM, ARIMA, Kalman Filtering, Data Assimilation, EEG/fMRI, Neural Encoding, Speech-Brain Modeling